Production Systems and Batching

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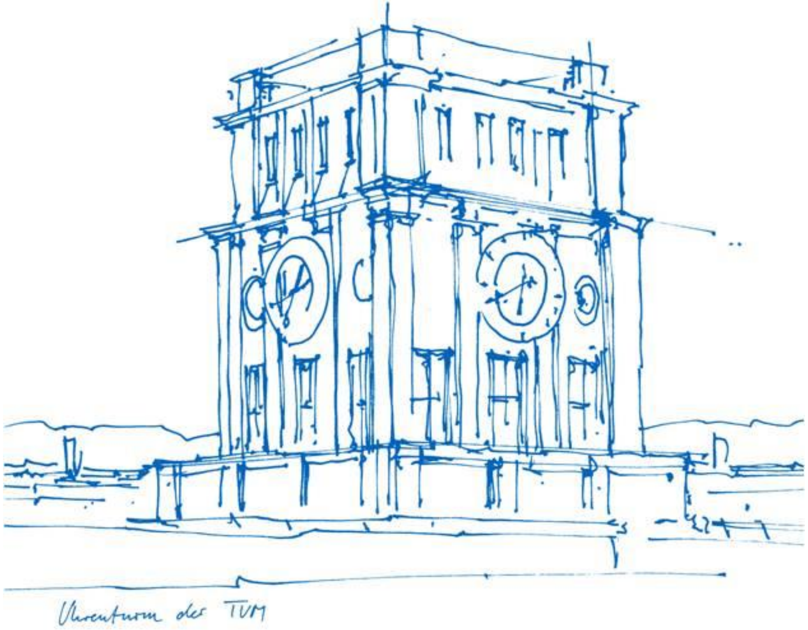


Production Systems

Production Systems and Applications

make-to-stock

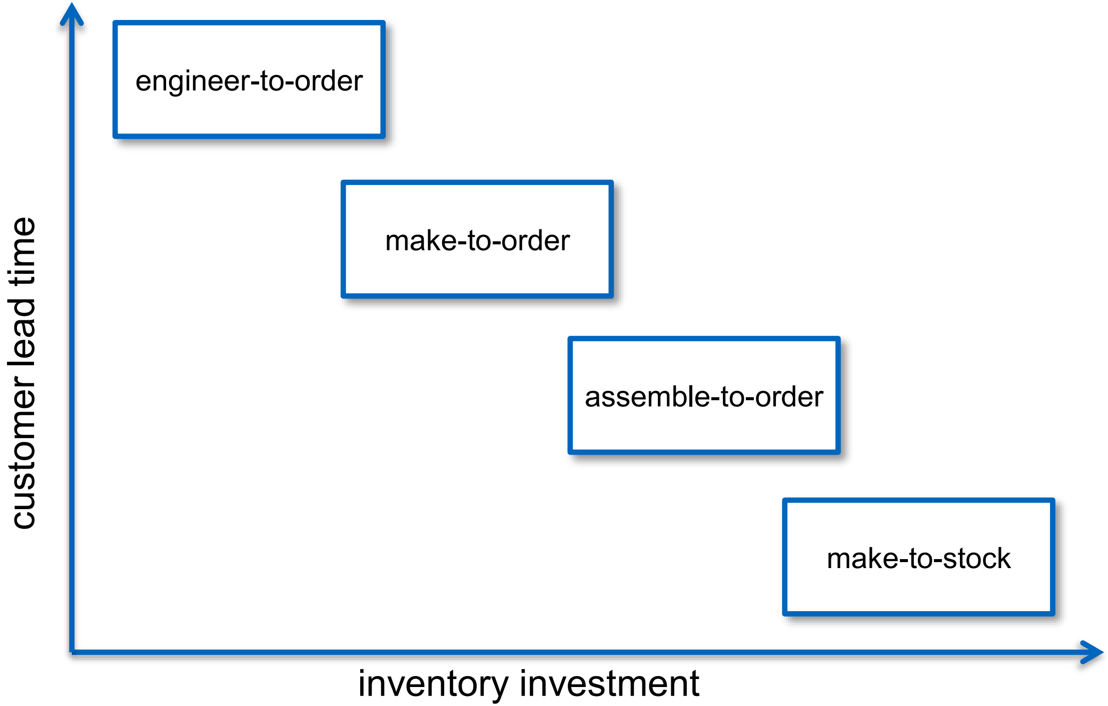
assemble-to-order

make-to-order

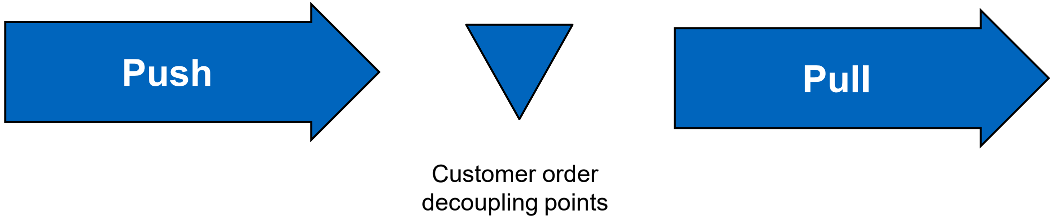
engineer-to-order

Source(s): *https://www.wikipedia.de/, https://pixabay.com/*

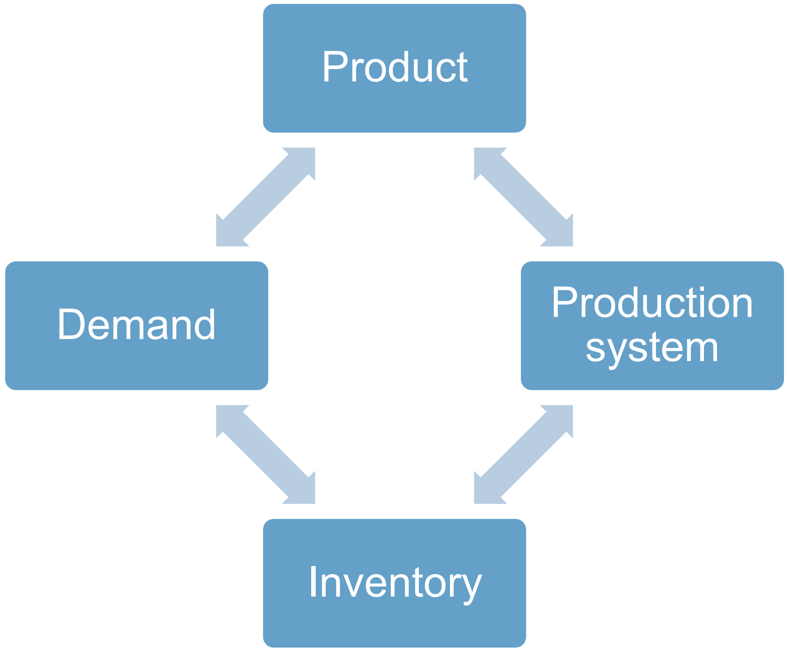
Positioning inventories



Push versus Pull Production

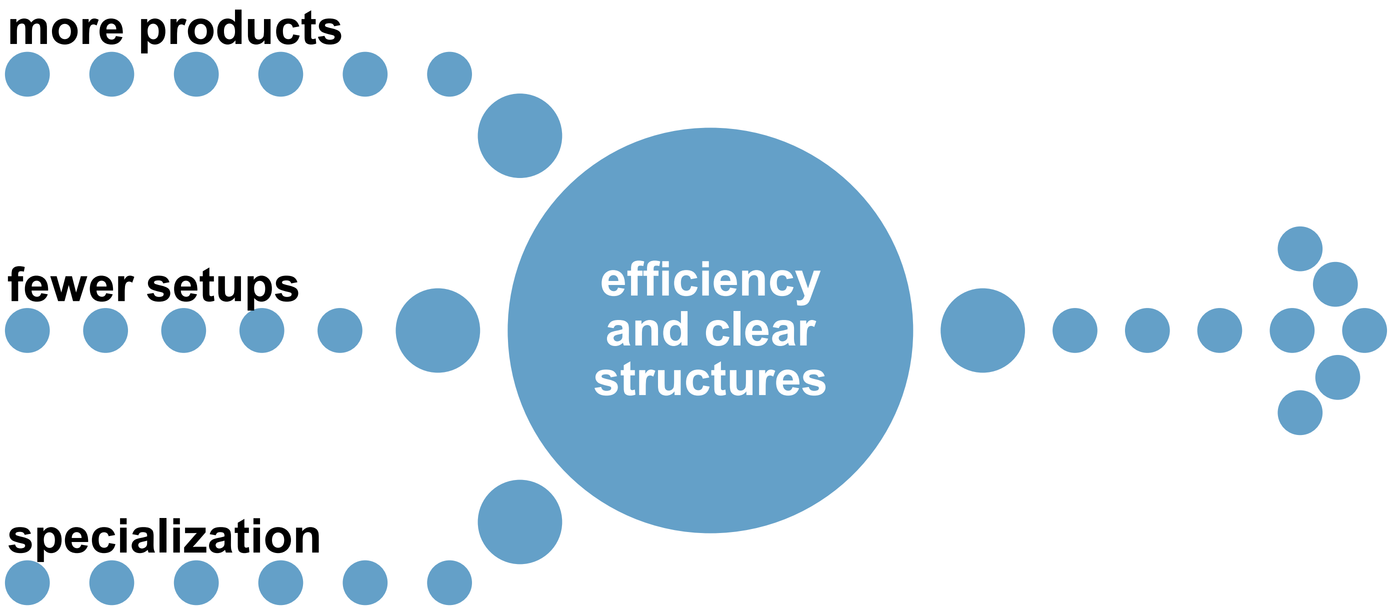


Production Systems and Their Context



Batch-and-Queue paradigm to Maximize Efficiency

Batch and Queue (Efficiency) Philosophy



Work Center Layout (Job Shop)

grouping of similar machines to departments

production of batches

setups between different batches, inventory

Batch-and-Queue

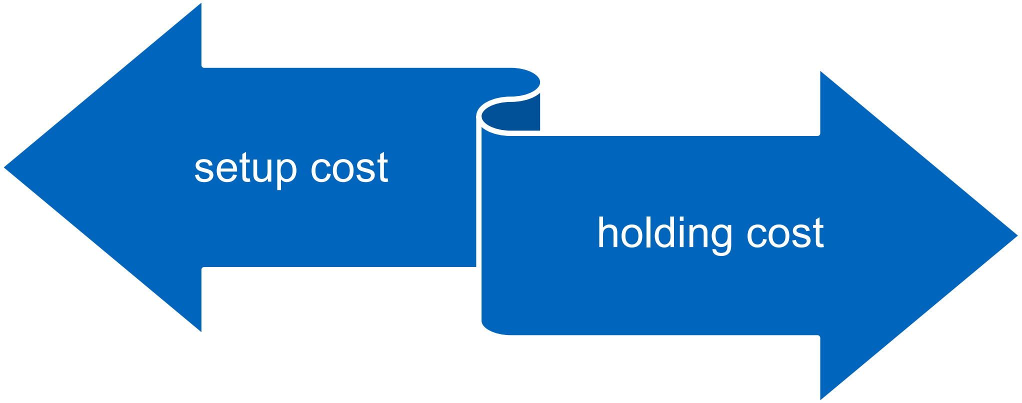
Economic Production Quantity Model

Deterministic Inventory and Production Management

To find the optimal number of batches (=setups) per year given deterministic and constant demand

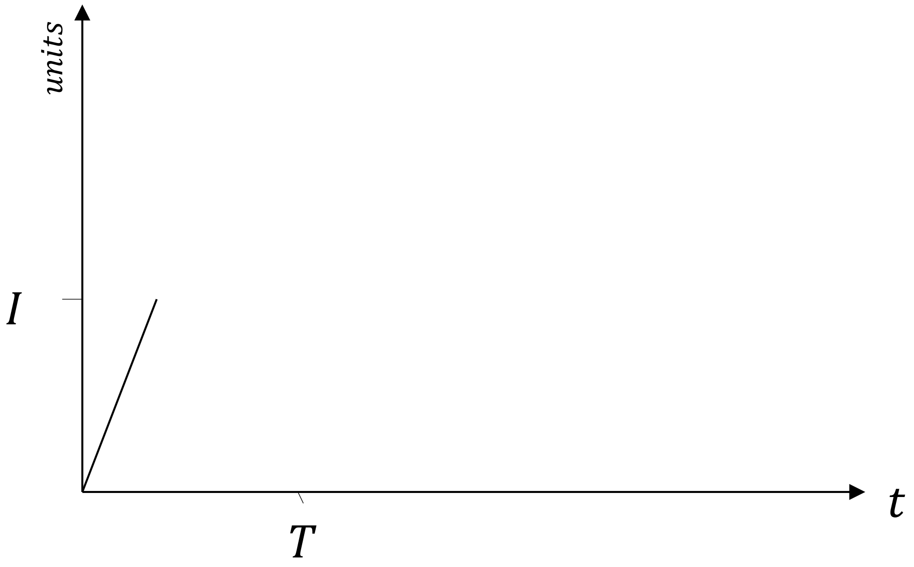
- only one product is involved
- the demand rate is constant and known
- shortages are not permitted
- there is no order lead time
- costs include
  - setup cost
  - holding cost
- **Production rate is finite**

Economic Production Quantities



What is the optimal batch size, Q?

Average Inventory Level



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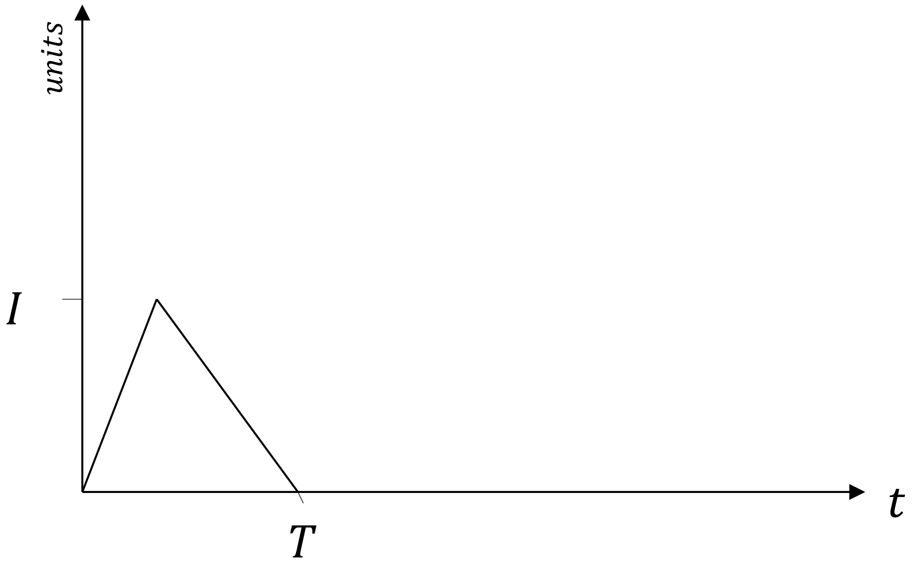

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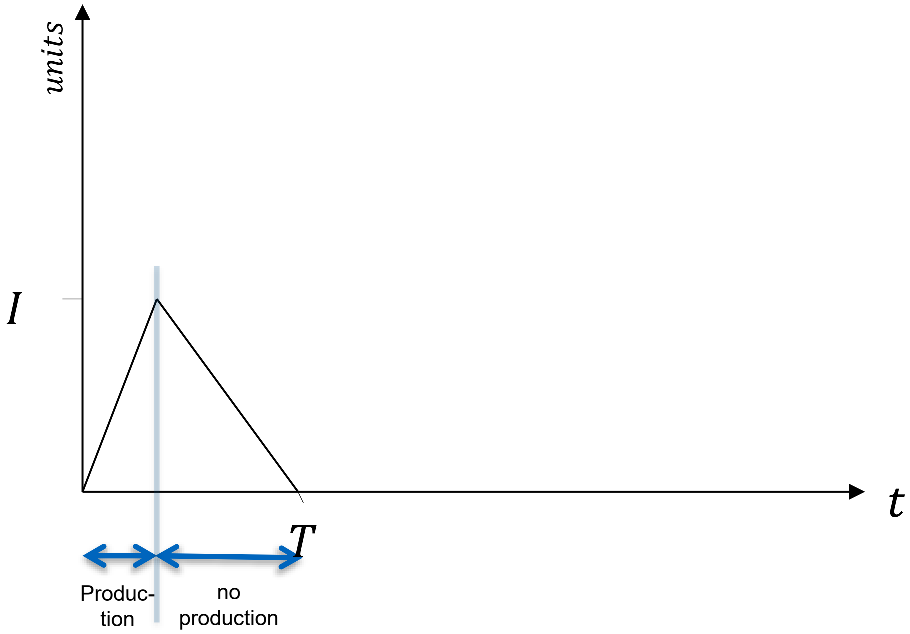

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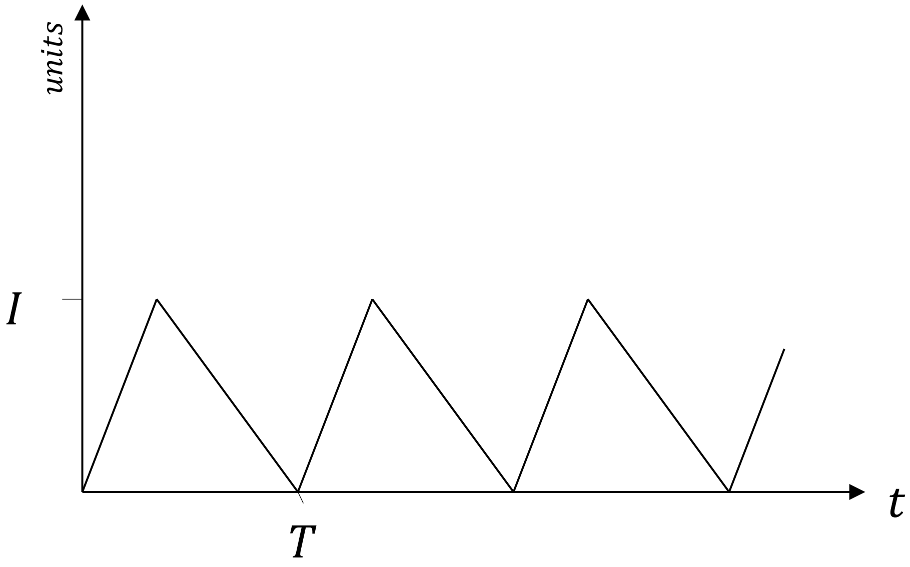

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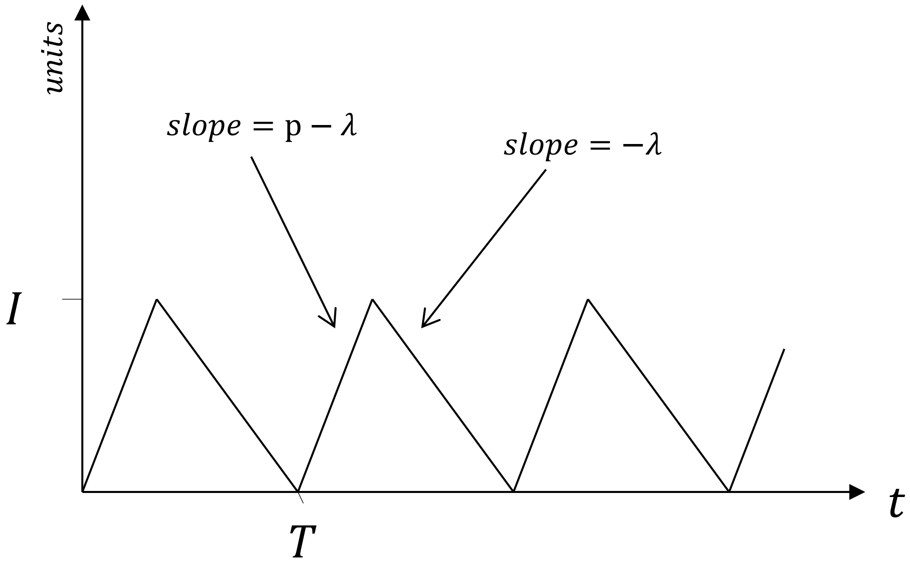

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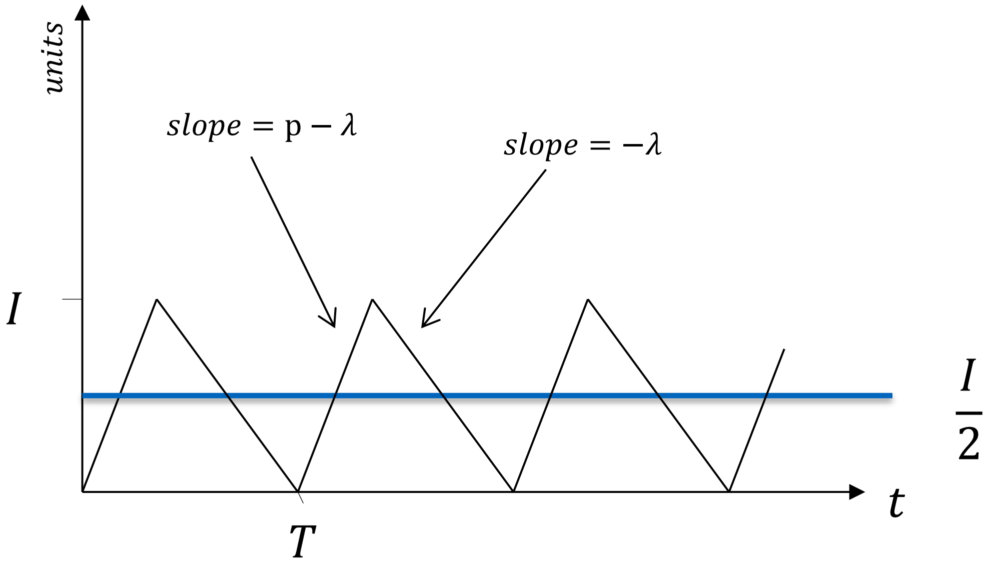
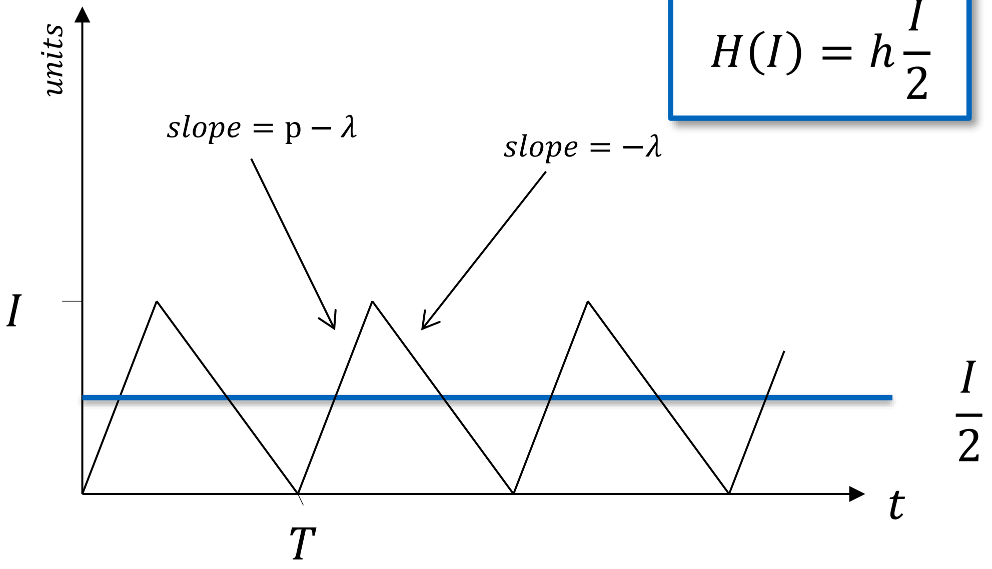
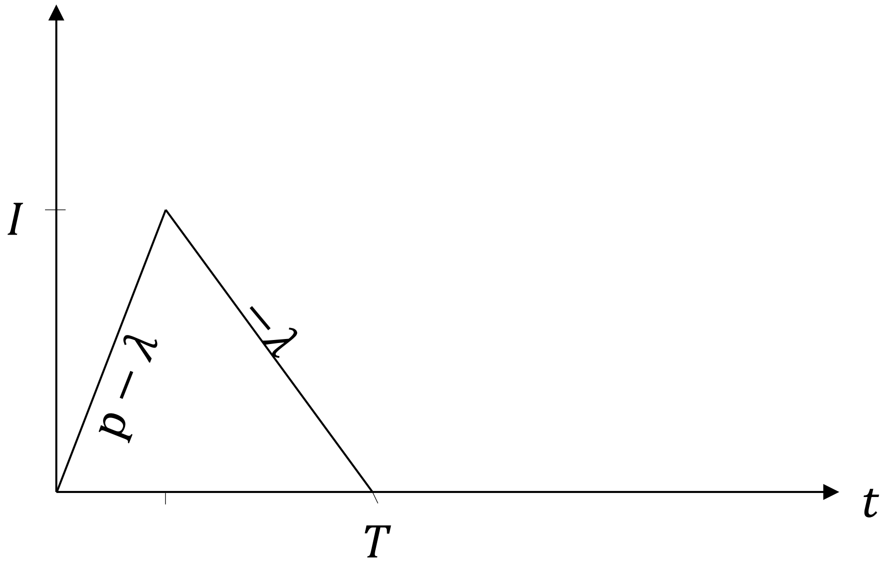


Average Inventory Level

$$\boxed{H(I) = h\frac{I}{2}}$$



Holding Costs as Function of Batch Size



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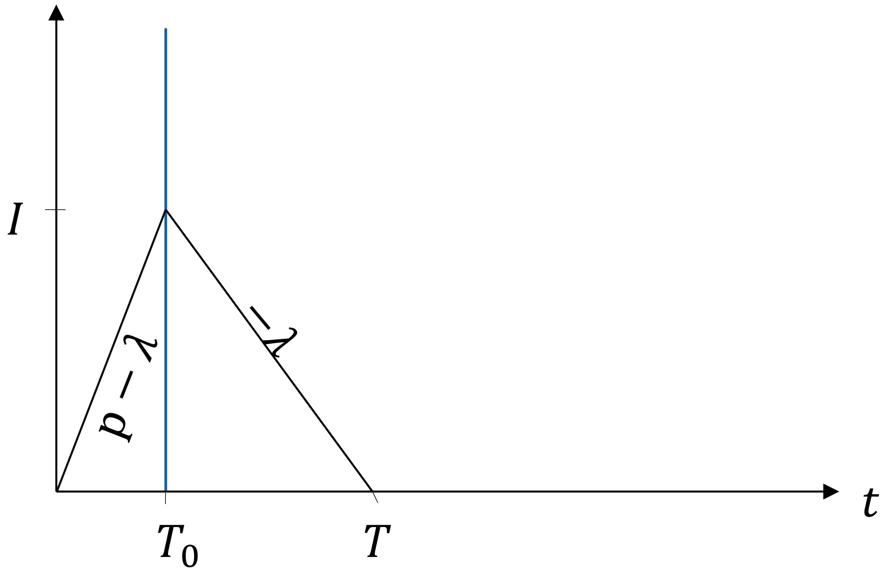

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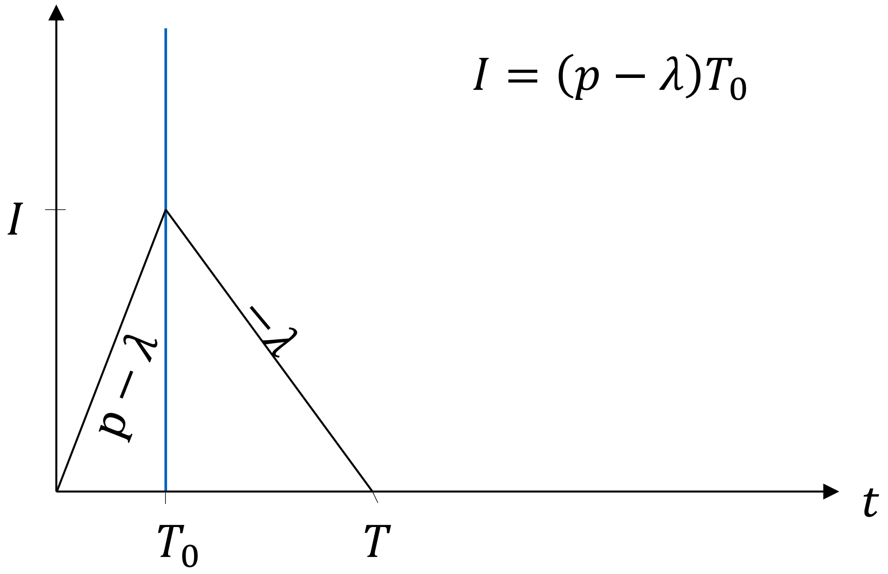
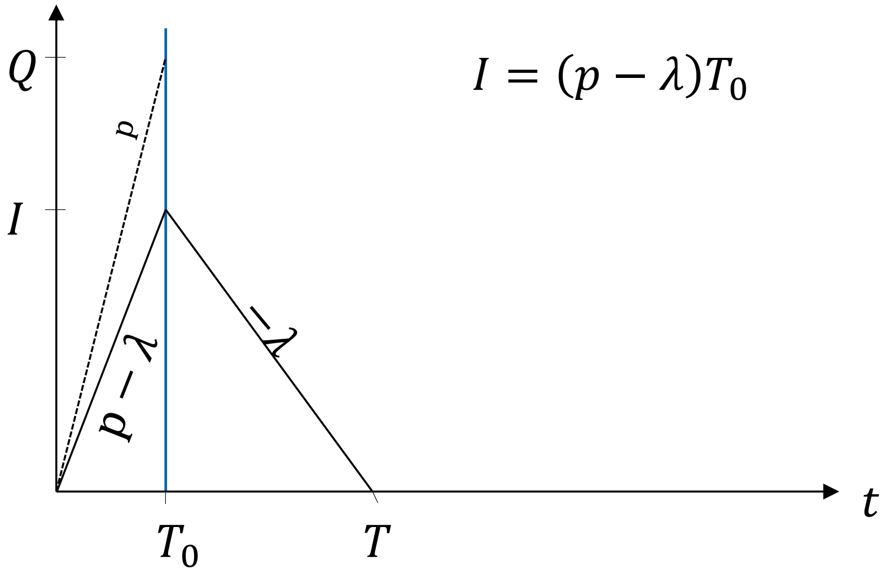


$$I = (p-\lambda)T_0$$

Holding Costs as Function of Batch Size

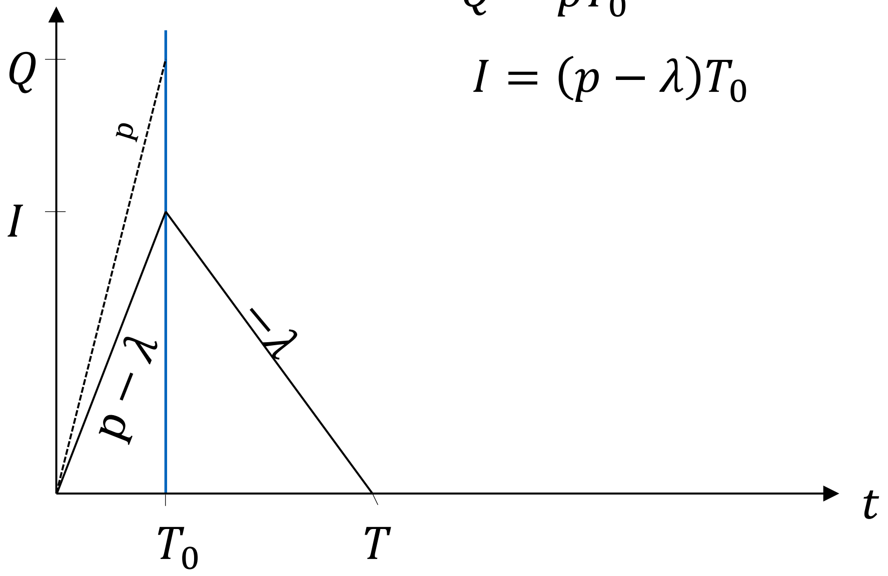


$$I = (p - \lambda)T_0$$

Holding Costs as Function of Batch Size

$$Q = pT_0$$

$$I = (p - \lambda)T_0$$



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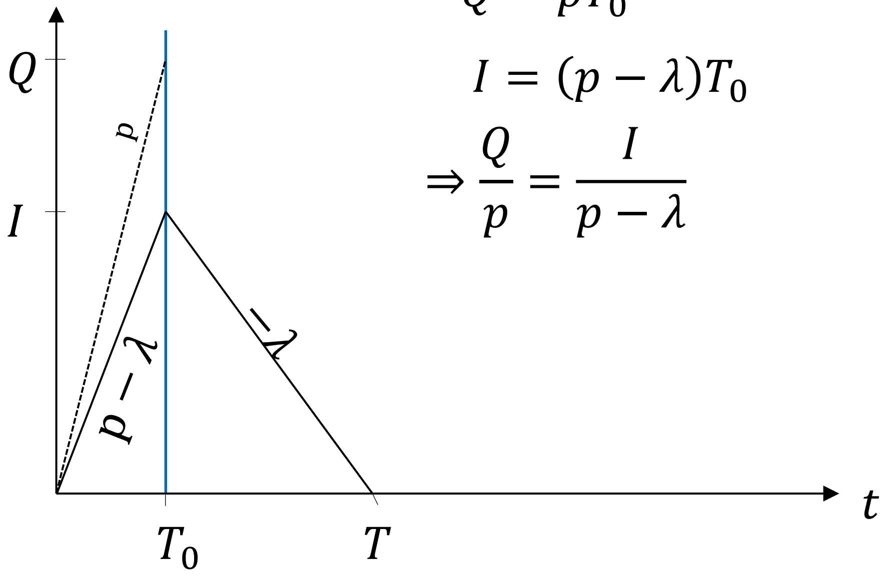

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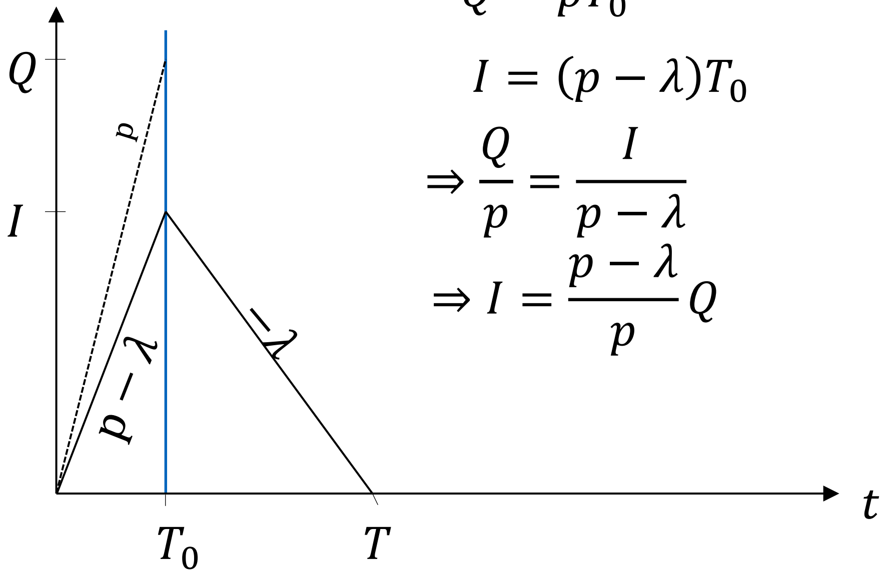
$$I = (p-\lambda)T_0$$

$$\Rightarrow \frac{Q}{p} = \frac{I}{p-\lambda}$$

Holding Costs as Function of Batch Size



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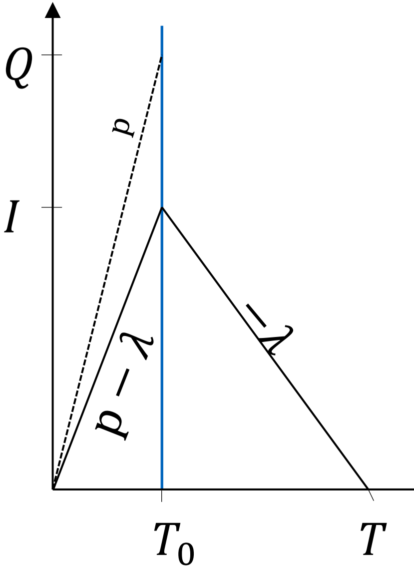
Holding Costs as Function of Batch Size



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$$I = (p-\lambda)T_0$$

$$\Rightarrow \frac{Q}{p} = \frac{I}{p-\lambda}$$

$$\Rightarrow I = \frac{p-\lambda}{p}Q$$

$$H(Q) = h\frac{p-\lambda}{p}\frac{Q}{2}$$

Average Number of Annual Setups

$$N = \frac{\lambda}{Q}$$

$$\boxed{k(Q) = K\frac{\lambda}{Q}}$$

Total Annual Costs

$$\boxed{TC(Q) = H(Q) + k(Q) = h\frac{p-\lambda}{p}\frac{Q}{2} + K\frac{\lambda}{Q}}$$

Optimal Order Quantity

$$TC'(Q) = \frac{p-\lambda}{p}\frac{h}{2} - K\frac{\lambda}{Q^2} := 0$$

$$Q^* = \sqrt{\frac{2K\lambda}{\frac{p-\lambda}{p}h}} = \sqrt{\frac{2K\lambda}{h}\frac{p}{p-\lambda}} = \sqrt{\frac{p}{p-\lambda}}\sqrt{\frac{2K\lambda}{h}}$$

$$TC''(Q) = 2K\frac{\lambda}{Q^3} > 0$$

Extreme Cases

$$Q^* = \sqrt{\frac{p}{p-\lambda}}\sqrt{\frac{2K\lambda}{h}}$$

$$p \to \infty:\ Q^* \to \sqrt{\frac{2K\lambda}{h}}, \qquad TC(Q^*) \to \sqrt{2hK\lambda}$$

$$p \to \lambda:\ Q^* \to \infty, \qquad TC(Q^*) \to 0$$

Duration of Production Runs

$$Q = pT_0$$

$$\Rightarrow T_0{}^* = \frac{Q^*}{p} = \sqrt{\frac{1}{p(p-\lambda)}}\sqrt{\frac{2K\lambda}{h}}$$

EPQ: Key Formulas

Key Formulas: Economic Production Quantity Model

$$TC(Q) = H(Q) + k(Q) = h\frac{p-\lambda}{p}\frac{Q}{2} + K\frac{\lambda}{Q}$$

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$$T_0{}^* = \frac{Q^*}{p} = \sqrt{\frac{1}{p(p-\lambda)}}\sqrt{\frac{2K\lambda}{h}}$$

Implications Matter More!